BIRLA INSTITUTE OF TECHNOLOGY, MESRA, RANCHI

COURSE STRUCTURE

B. Sc. (BUSINESS ANALYTICS)

as per NEP-2020

w.e.f. Academic Year 2025-26

Total Credit Requirement for B. Sc. (Business Analytics) = 123 (in 3 years/ 6 semesters)

Total Credit Requirement for B. Sc. Honours (Business Analytics) = 163 (in 4 years/ 8 semesters)

FIRST MONSOON / SEMESTER 1

Semester/Session	Level	Category	Courses	L	T	P	C
First/Monsoon	FIRST	AECC	HS24131 - Communication Skills - I	0	0	3	1.5
		MDC	MN25118 - Introduction to Computing	3	0	0	3
		VAC	MN25102 - Human Values & Professional Ethics	2	0	0	2
		VAC	VAC - 1 (One Course from VAC Group) *				2
		SEC-SB	CA25110 - Office Automation Tools	0	1	4	3
		DSC	BA25101 - Principles and Practices of Management	3	0	0	3
		DSC	MN25119 - Statistics for Managers	3	0	0	3
		DSC	MN25108 - Introduction to Business Accounting	3	0	0	3
TOTAL CREDITS							20.5

SECOND SPRING / SEMESTER 2

Semester/Session	Level	Category	Courses	L	T	P	C
Second/Spring	FIRST	AECC	MN25109 - Public Speaking & Creative Writing	1	0	2	2
		MDC	BA25102 - Python Programming	3	0	3	4.5
		VAC	VAC - 2 (One Course from VAC Group)*				2
		SEC-SB	BA25103 - Advanced Microsoft Excel	0	1	4	3
		DSC	BA25104 - Marketing Management	3	1	0	4
		DSC	MN25115 - Business Economics	3	0	0	3
		DSC	BA25105 - Fundamentals of Business Analytics	3	0	0	3
TOTAL CREDITS							21.5

EXIT OPTION AFTER FIRST YEAR WITH CERTIFICATION

Students exercising exit option after first year have to pick up two courses of 3 credits each (total 6 credits) from the Vocational Course list (level-I). For others, the vocational courses are optional and non-credit.

THIRD MONSOON / SEMESTER 3

Semester/Session	Level	Category	Courses	L	T	P	C
Third/Monsoon	SECOND	AECC	MT24204 - Personality Development	2	0	2	3
		MDC	BA25201 - R Programming	3	0	3	4.5
		SEC-SB	BA25202 - Database Management System	2	0	2	3
		DSC	BA25203 - Business Intelligence and Analytics	3	1	0	4
		DSC	BA25204 - Quantitative Analysis for Business	3	0	0	3
		DSE	DSE Elective - I	3	0	0	3
TOTAL CREDITS							20.5

FOURTH SPRING / SEMESTER 4

Semester/Session	Level	Category	Courses	L	T	P	C
Fourth/Spring	SECOND	AECC	HS24133 - Communication Skills - II	0	0	3	1.5
		MDC	BA25205 - Big Data Technologies	3	0	3	4.5
		SEC-SB	BA25206 - SQL for Data Analytics	2	0	2	3
		DSC	BA25207 - Statistical Methods and Inference	3	1	0	4
		DSC	BA25208 - Financial Analytics	3	0	0	3
		DSE	DSE Elective - II	3	0	0	3
TOTAL CREDITS							19.5

EXIT OPTION AFTER SECOND YEAR WITH DIPLOMA

Students exercising exit option after second year have to pick up two courses of 3 credits each (total 6 credits) from the Vocational Course list (level-II). For others, the vocational courses are optional and non-credit.

FIFTH MONSOON / SEMESTER 5

Semester/Session	Level	Category	Courses	L	T	P	C
Fifth/Monsoon	THIRD	DSC	BA25301 - Machine Learning for Analytics	3	0	2	4
		DSC	BA25302 - Advanced Data Visualization	2	0	2	3

		DSC	BA25303 - Predictive Analytics	3	1	0	4
		DSC	BA25304 - Market Research and Consumer Analytics	3	0	0	3
		DSE	DSE Elective - III	3	0	0	3
TOTAL CREDITS							17

SIXTH SPRING / SEMESTER 6

Semester/Session	Level	Category	Courses	L	T	P	C
Sixth/Spring	THIRD	DSC	BA25305 - Text Analytics and NLP	3	0	2	4
		DSC	BA25306 - Time Series Analysis	3	1	0	4
		DSC	BA25307 - Network and Social Media Analytics	2	0	2	3
		DSE	DSE Elective - IV	3	0	0	3
		DSE	DSE Elective - V	3	0	0	3
TOTAL CREDITS							17

EXIT OPTION AFTER THIRD YEAR WITH DEGREE

Total Credits for B. Sc. (Business Analytics) = 123

SEVENTH MONSOON / SEMESTER 7

Semester/Session	Level	Category	Courses	L	T	P	C
Seventh/Monsoon	FOURTH	DSC	BA25401 - Deep Learning and AI for Analytics	3	0	2	4
		DSC	BA25402 - Business Strategy and Analytics	3	1	0	4
		DSE	DSE Elective - VI	3	0	0	3
		DSE	DSE Elective - VII	3	0	0	3
		Research/Project	BA25403 - Research Internship/Project	0	0	0	12
TOTAL CREDITS							26

EIGHTH SPRING / SEMESTER 8

Semester/Session	Level	Category	Courses	L	T	P	C
Eighth/Spring	FOURTH	DSC	BA25404 - Advanced Analytics Capstone	2	0	4	4
		DSE	DSE Elective - VIII	3	0	0	3

		Research/Project	BA25405 - Major Research Project/Dissertation	0	0	0	20
TOTAL CREDITS							27

B. Sc. HONOURS (BUSINESS ANALYTICS)

Total Credits for B. Sc. Honours (Business Analytics) = 163

ELECTIVE AND VOCATIONAL COURSES

VAC Electives (Value Added Courses)

VAC Elective	Code	Course Name	L	T	P	C
VAC-1/VAC-2	MN25103	Yoga	1	0	2	2
	MN25104	Physical Education	1	0	2	2
	MN25111	Digital Empowerment	2	0	0	2
	MN25112	Emotional Intelligence	2	0	0	2

DSC Electives (Discipline Specific Electives)

DSE Elective	Code	Course Name	L	T	P	C
DSE-I/II/III	BA25209	Digital Marketing Analytics	3	0	0	3
	BA25210	Operational Analytics	3	0	0	3
	BA25211	Supply Chain Analytics	3	0	0	3
	BA25212	Risk Analytics	3	0	0	3
DSE-IV/V/VI	BA25308	Healthcare Analytics	3	0	0	3
	BA25309	Sports Analytics	3	0	0	3
	BA25310	IoT and Sensor Data Analytics	3	0	0	3
DSE-VII/VIII	BA25311	Cloud Analytics	3	0	0	3
	BA25312	Computer Vision and Image Analytics	3	0	0	3

Vocational Courses - Level I

Course Code	Course Name	L	T	P	C
BA25501	Data Entry and Basic Operations	2	0	4	3
BA25502	Spreadsheet Skills for Analytics	2	0	4	3
BA25503	Introduction to Data Warehousing	2	0	4	3
BA25504	Web Analytics Fundamentals	2	0	4	3

Vocational Courses - Level II

Course Code	Course Name	L	T	P	C
BA25505	Tableau for Business Analytics	2	0	4	3
BA25506	Power BI for Analytics	2	0	4	3
BA25507	Google Analytics and Marketing Analytics	2	0	4	3
BA25508	Excel Advanced Analytics	2	0	4	3

Vocational Courses - Level III

Course Code	Course Name	L	T	P	C
BA25509	Hadoop and Spark for Big Data	2	0	4	3
BA25510	Advanced Python for Analytics	2	0	4	3
BA25511	Certification Preparation - Google Analytics	2	0	4	3
BA25512	Case Studies in Business Analytics	2	0	4	3

ACRONYMS EXPANDED

AECC: Ability Enhancement Compulsory Course

MDC: Multidisciplinary Course

VAC: Value Added Course

SEC-SB: Skill Enhancement Course-Skill Based

DSC: Discipline Specific Core (Course)

DSE: Discipline Specific Elective (Course)